

Designing fraud-prevention systems that keep analysts in the loop

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Abstract. This article extends the Nerdearla Chile 2026 talk on fraud prevention, machine learning, and design patterns. The main claim is that fraud prevention should be treated as an **ML-enabled software-architecture problem**, not only as a classification problem. The article develops a design-pattern proposition for **ML-enabled Human-in-the-Loop (HIL) Triage**: rules for explicit cases, ML for scoring, policy for routing, and expert analysts for ambiguous cases and structured feedback. The discussion connects this architecture to the literature on software architecture for ML systems, human-in-the-loop design, design patterns for AI-based systems, learning to defer, alert prioritization, reliable ML, and platform engineering ([Mosqueira-Rey et al., 2023](#)).

1 Introduction

This article fills the gap left by a short conference talk: the broader software-architecture background, the design-pattern framing, the operational consequences, and the connections to MLOps and platform engineering.

The central thesis: robust fraud-prevention systems are hybrid systems. They rarely succeed as purely manual or purely automatic workflows. The most resilient architecture combines deterministic controls, statistical scoring, decision policy, and expert human judgment.

That claim is architectural. Production ML systems require supporting software, data pipelines, monitoring, deployment structures, and cross-functional engineering practices ([Amershi et al., 2019a](#); [Lewis et al., 2021b](#); [Nazir et al., 2024](#); [Sculley et al., 2015](#)). In fraud prevention, the surrounding system also includes analysts, decision queues, escalation paths, and operational feedback loops. The unit of design is the socio-technical system ([Dobbe and Wolters, 2024](#); [Salwei and Carayon, 2022](#)).

Hybrid system:

Combines deterministic rules, ML scoring, decision policy, and expert human judgment rather than relying on any single component.

Socio-technical system:

Behavior shaped by both technical components (models, pipelines, infrastructure) and human work (analyst review, organizational decisions, governance).



Figure 1. Effective fraud systems are built from the interaction of rules, ML, and expert analysts.

Figure 1 captures the argument in its simplest form. Rules encode explicit knowledge. ML compresses signals into scores. Analysts contribute judgment, novelty detection, and feedback. The point is to design their interaction well.

2 Why fraud prevention becomes an architecture problem

Fraud prevention is not a stable prediction problem. Systems combine rules, supervised models, anomaly detection, and human investigation because no single technique handles novelty, delayed labels, and operational constraints (Carcillo et al., 2021; Hernandez Aros et al., 2024).

First, attackers adapt (Bolton and Hand, 2002; Carcillo et al., 2021; Lunghi et al., 2023).

Second, labels are delayed and imperfect (Carcillo et al., 2021; Lunghi et al., 2023). The system acts inside the environment that later generates its own labels.

Third, error costs are asymmetric (Bolton and Hand, 2002; Hernandez Aros et al., 2024).

Fourth, many decisions are time-sensitive (Bolton and Hand, 2002; Carcillo et al., 2021; Lunghi et al., 2023). Architecture determines which cases are handled automatically and how scarce analyst attention is reserved for the ambiguous zone.

These properties explain the move toward system-centric and architecture-centric thinking (Lewis et al., 2021b; Muccini and Vaidhyanathan, 2021; Nazir et al., 2024).

Adversarial adaptation: Fraudsters change tactics once a defense becomes known, making detection

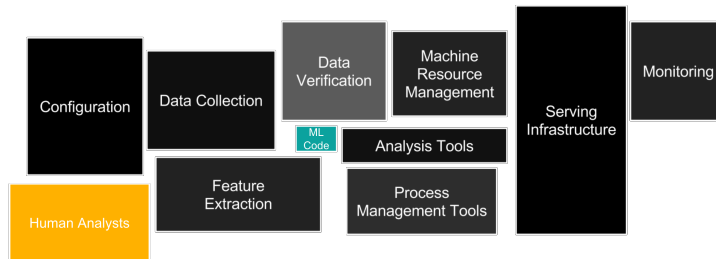
Label delay: Ground truth is often confirmed only after complaints or investigations—

Asymmetric error costs: False negatives cause direct loss; false positives

Time sensitivity: A correct decision made too late may still be operationally poor.

3 From hidden technical debt to ML-enabled fraud systems

Sculley et al. (2015) argued that much of the real cost of ML systems lies outside the model itself. A fraud model is never the whole fraud system.



Hidden technical debt: The real cost of production ML lies mostly outside the model: data pipelines, configuration, serving, monitoring, and glue code.

Figure 2. System anatomy for fraud operations. Adapted from Sculley et al. (2015).

Figure 2 places the model inside a wider environment. Lin and Ryaboy (2013) reinforce this from large-scale data-mining infrastructure.

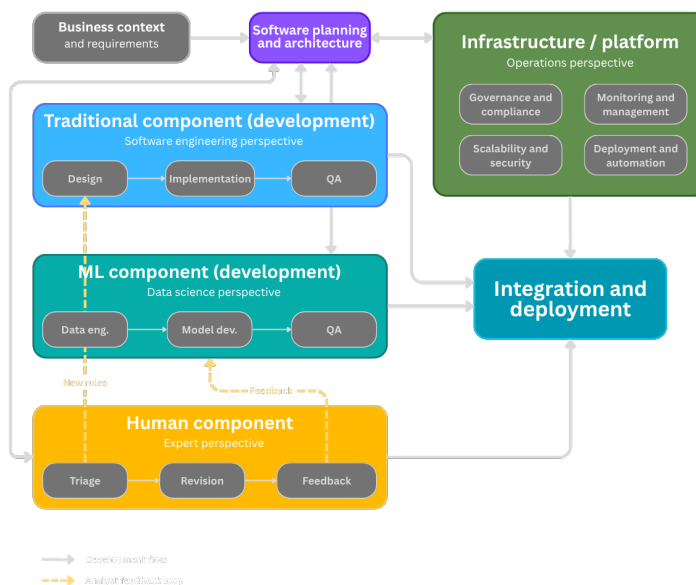


Figure 3. Planning view. Adapted from Lewis et al. (2021b), Andersen and Maalej (2024), and Kästner (2025).

Figure 3 broadens the lens to planning. Lewis et al. (2021b) motivate co-architecting and co-versioning. The added human lane makes explicit what fraud operations often leave implicit.

Co-versioning: Data, features, models, evaluation artifacts, and policy thresholds should be traceable together.

4 Why this article proposes a design pattern

The architecture is proposed as a reusable response to a recurring class of problems.

Design pattern: A reusable solution to a recurring problem. Specifies context, forces,

ML architecture patterns: Washizaki et al. identified 33 ML patterns. Heiland et al. expanded to 70.

Washizaki et al. (2020) documented recurring ML patterns. Heiland et al. (2023) expanded the repository. Järvenpää et al. (2024) focused on reusable tactics. Cruz et al. (2023) reinforced architecture rationale.

Andersen and Maalej (2024) frame HIL as a design space with reusable patterns.

Mosqueira-Rey et al. (2023) describe HIL-ML as a family of approaches.

HIL patterns: Andersen & Maalej catalog reusable HIL patterns spanning

HIL-ML family: Active learning, interactive ML, machine teaching. Differs in *how* and *when* human expertise enters the loop.

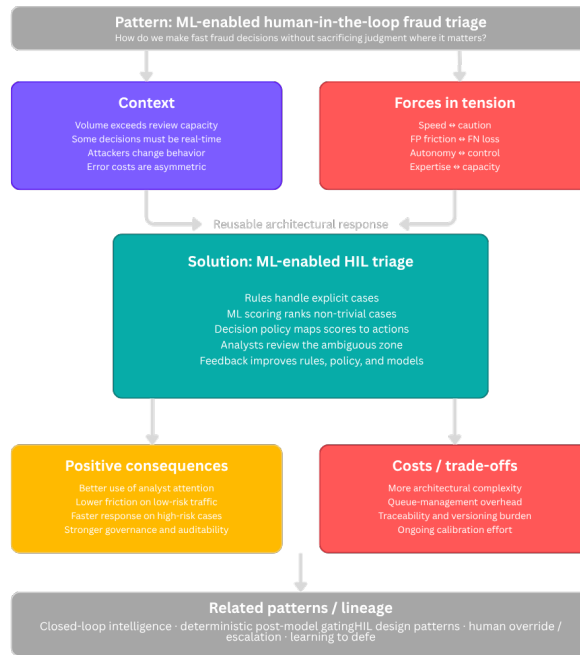
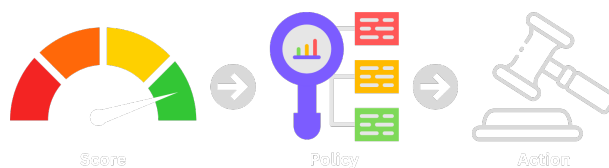


Figure 4. Pattern framing for ML-enabled HIL fraud triage. Synthesized from Lakshmanan et al. (2020), Andersen and Maalej (2024), Heiland et al. (2023), Cruz et al. (2023), Järvenpää et al. (2024).

Figure 4 summarizes the pattern view. Forces and consequences make trade-offs explicit, which is central to architectural reasoning (Richards and Ford, 2025).

5 The anatomy of the proposed pattern

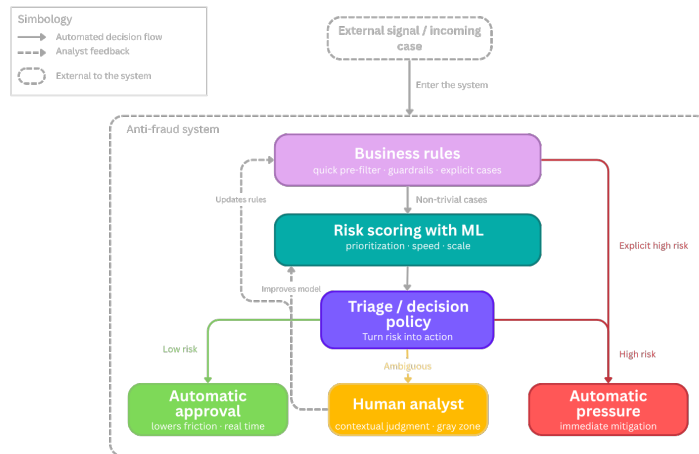
The pattern is a series of transformations: observation to score, score to policy, policy to action.



Score $\neq$ decision: The model produces a risk estimate. A separate policy layer translates it into action under business constraints.

Figure 5. Score-to-policy-to-action pipeline.

The separation in Figure 5 avoids treating the score as a business decision (Chen et al., 2022; Kästner, 2025).



Learning to defer (L2D):

The system decides whether to handle a case automatically or route it to a human expert.

Figure 6. Reference architecture of the ML-enabled HIL Triage pattern.

Figure 6 summarizes the runtime logic. This aligns with the learning-to-defer literature (Alves et al., 2025).

6 Analysts are a runtime component, not a fallback afterthought

Analysts should be explicit runtime components.



Analyst as component:

Designed workflow with structured inputs (ranked cases, reason codes) and structured outputs (decisions, feedback).

Figure 7. Bidirectional collaboration between analysts and ML.

Figure 7 shows two-way collaboration (Ding et al., 2023; Mosqueira-Rey et al., 2023).

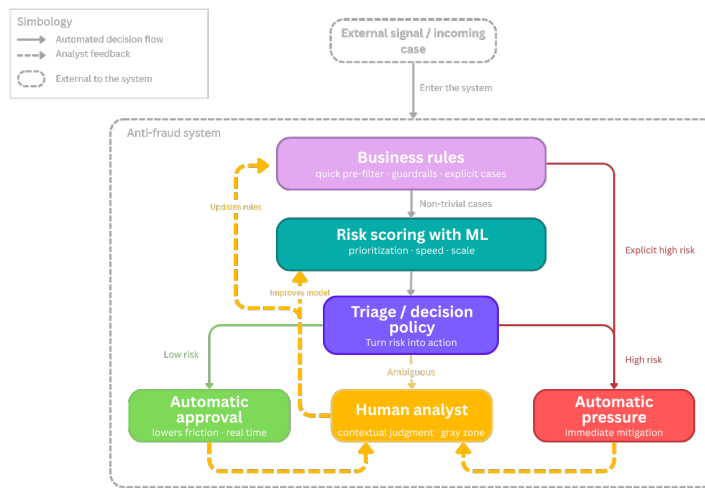


Figure 8. Closed-loop HIL triage with explicit feedback paths.

Kadam (2024) treats feedback as propagable and reusable signal.

Expert-in-the-loop approaches suggest human review is most valuable where the model faces novelty (Yuan et al., 2026).

Analysts should also audit automatically resolved cases (Andersen and Maalej, 2024; Chen et al., 2022; Kadam, 2024; Tan Wei Hao et al., 2026).

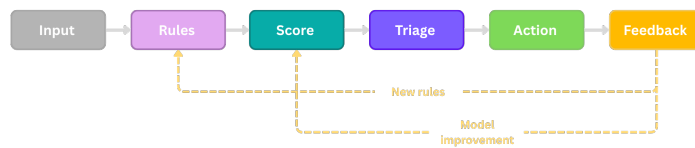


Figure 9. Operative learning: the pattern is a loop, not a pipeline.

Feedback propagation:

Even sparse analyst annotations can improve

Open-set recognition:

Detecting cases outside any known class. Human

Retrospective audit:

Review samples of automatically resolved cases to catch errors and revise routing logic.

Fraud analyst workbench 12 pending

CASE	BAND	REASON CODES	SLA	
TXN-90281 \$4,828 - new device - BR	High	velocity geo-shift	0:12:41	Review
TXN-90277 \$1,205 - known device - MX	Ambiguous	amount-spike	0:34:18	Review
TXN-90264 \$789 - returning user - CL	Ambiguous	pattern-break time-of-day	0:48:02	Review
TXN-90259 \$2,188 - new account - AR	Ambiguous	new-act high-value	0:55:30	Review
TXN-90252 \$348 - known device - CO	Ambiguous	frequency	1:02:11	Review
TXN-90248 \$615 - known device - PE	Ambiguous	merchant-cat	1:14:55	Review

Case detail: TXN-90281

RISK SCORE 0.87

TOP SIGNALS

- Device age: 2 hrs
- Txn velocity (1h): 6 txns
- Geo distance: 4,200 km
- Account age: 14 days
- Amount percentile: P98

SLA STATUS

12 min 41 sec (SLA: 30 min)

ANALYST DECISION

Approve Block Escalate

STRUCTURED FEEDBACK

Reason code...
Confidence...
Escalation path...

Analyst workbench:

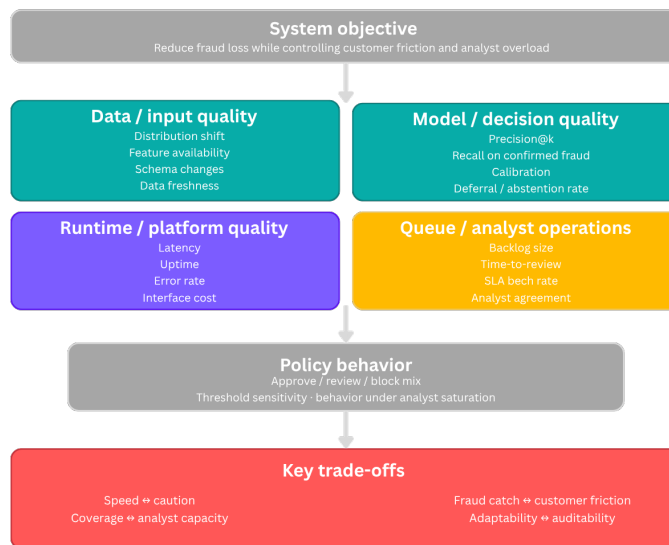
Ranked cases, risk bands, reason codes, SLA timers, action controls, structured feedback fields.

Figure 10. Analyst workbench. Synthesized from Jalalvand et al. (2024), Ghadermazi et al. (2024), Alves et al. (2025).

Figure 10 turns the architecture into an operational review system (Alves et al., 2025; Amershi et al., 2019b; Ghadermazi et al., 2024; Jalalvand et al., 2024; Zelvenskiy et al., 2022).

7 Evaluation must fit the socio-technical system

Evaluation has to widen (Dobbe and Wolters, 2024; Lewis et al., 2021b; Nazir et al., 2024; Salwei and Carayon, 2022).



Four evaluation layers:
(1) Data/input quality,
(2) model/decision quality, (3) queue/analyst operations,
(4) runtime/platform quality.

Figure 11. Metrics and trade-offs for ML-enabled HIL fraud triage.

Figure 11 defines a broader evaluation frame (Alves et al., 2025; Chen et al., 2022; Dobbe and Wolters, 2024; Fazelpour, 2025; Ghadermazi et al., 2024; Jalalvand et al., 2024; Lewis et al., 2021b; Tan Wei Hao et al., 2026).

The policy box is equally important (Chen et al., 2022; Kästner, 2025; Lewis et al., 2021b). The trade-off strip reflects operational frontiers (Chen et al., 2022; Fazelpour, 2025; Richards and Ford, 2025).

Deferral rate: Fraction of cases routed to human review. Key tuning parameter for workload.

Policy behavior: The approve/review/block split is an organizational choice, not just a model output.

8 MLOps and platform engineering are part of the argument

The pattern has an operational lifecycle (Kästner, 2025; Lewis et al., 2021b; Tan Wei Hao et al., 2026).

MLOps lifecycle: Data → features → training → eval → deploy → monitor → review → feedback. Three return loops.

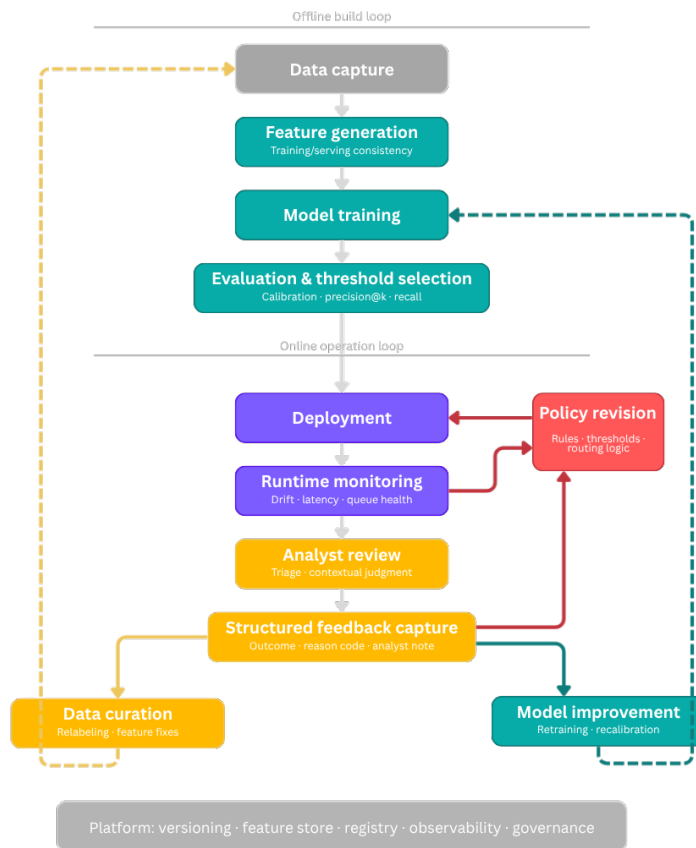


Figure 12. MLOps feedback lifecycle. Adapted from Lewis et al. (2021b), Tan Wei Hao et al. (2026), Kadam (2024).

The side loops show why feedback should not be collapsed (Kadam, 2024; Konieczny, 2025; Lewis et al., 2021a,b; Lin and Ryaboy, 2013; Mosqueira-Rey et al., 2023).

Versioning, feature storage, registry services, observability, and governance are part of what makes the architecture maintainable.

9 What the pattern improves, and what it costs

The pattern improves analyst efficiency (Alves et al., 2025; Jalalvand et al., 2024), reduces friction (Jalalvand et al., 2024; Kästner, 2025), separates inference from governance (Chen et al., 2022; Lewis et al., 2021b), and makes learning explicit (Andersen and Maalej, 2024; Mosqueira-Rey et al., 2023). Costs include more moving parts (Lewis et al., 2021b; Nazir et al., 2024), queue monitoring (Ghadermazi et al., 2024; Jalalvand et al., 2024), versioned decision logic (Cruz et al., 2023; Lewis et al., 2021b), and cross-team coordination (Nazir et al., 2024; Richards and Ford, 2025).

9.1 Anti-patterns worth naming

Monitorability:
First-class quality attribute. Separate monitoring for data,

ML platform: Versioning, feature store, model registry, observability, governance. Reduces accidental complexity.

Anti-pattern — Score = policy: Raw scores trigger actions; governance becomes opaque.

Letting the score become policy (Lewis et al., 2021b; Washizaki et al., 2020). Remedy: Section 5 (Kästner, 2025; Washizaki et al., 2020).

Treating analysts as unstructured exception handling (Andersen and Maalej, 2024; Mosqueira-Rey et al., 2023).

Building the model while leaving operations underspecified (Jalalvand et al., 2024; Lewis et al., 2021b; Nazir et al., 2024).

Ignoring monitorability until something breaks (Jalalvand et al., 2024; Lewis et al., 2021b).

Glue code and pipeline jungles (Kästner, 2025; Lewis et al., 2021b; Washizaki et al., 2020). These anti-patterns define the pattern's limits (Cruz et al., 2023; Richards and Ford, 2025).

Anti-pattern — Unstructured HIL: Free-text-only review generates cost but no reusable signal.

Anti-pattern — Pipeline jungle: Feature joins, rule engines, risk services accumulate without platform discipline.

10 Why this matters for a new, but important field

ML-enabled software architecture is still young (Muccini and Vaidhyathan, 2021; Nazir et al., 2024). Human participation is itself a design space (Andersen and Maalej, 2024; Mosqueira-Rey et al., 2023). Fraud makes the socio-technical nature of ML visible (Dobbe and Wolters, 2024; Kästner, 2025; Salwei and Carayon, 2022). This pattern adds one concrete example to that growing field (Andersen and Maalej, 2024; Kästner, 2025; Richards and Ford, 2025; Washizaki et al., 2020).

11 Open questions

One concerns **policy optimization**. Another concerns **feedback quality** (Alves et al., 2025; Yuan et al., 2026). A third concerns **architecture evaluation** (Chen et al., 2022; Cruz et al., 2023; Jalalvand et al., 2024; Lunghi et al., 2023; Nazir et al., 2024). A fourth concerns **platform productization** (Lin and Ryaboy, 2013; Tan Wei Hao et al., 2026).

A fifth concerns **LLMs in the analyst workflow**: how to integrate case summarization, reasoning traces, and explanation interfaces without weakening governance (Tan Wei Hao et al., 2026).

LLM integration: How to add NL case summarization and conversational investigation without undermining structured feedback and auditability.

12 Conclusion

Fraud prevention should be designed as an ML-enabled socio-technical architecture. Use rules where explicit knowledge exists. Use ML where prioritization creates scale. Use analysts where context and judgment matter. Keep score separate from policy. Capture feedback in structured form. Design the platform so the system can be observed, revised, and improved.

This combination of rules, scoring, policy, analyst review, and closed-loop learning deserves to be treated as a reusable architectural response for a

high-stakes domain.

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